



# *Computer Vision for Autonomous Vehicles*

## *Quadrotor Control*

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*University of Oklahoma*

*Fall 2025*

*November 2025*



# Announcement

Lab 7 (optional) will be posted this week: due date: Friday Dec 19<sup>th</sup>, 11:59 PM  
(no extension is accepted)

Final project demonstration and presentation: Dec 16<sup>th</sup>, 1:30-3:30 in REPF B4

Final project report due: Thursday, Dec 18, 11:59 PM

Final exam (take-home) deadline: Sunday December 21<sup>st</sup>, 11:59 PM (no  
extension is accepted)

# Today

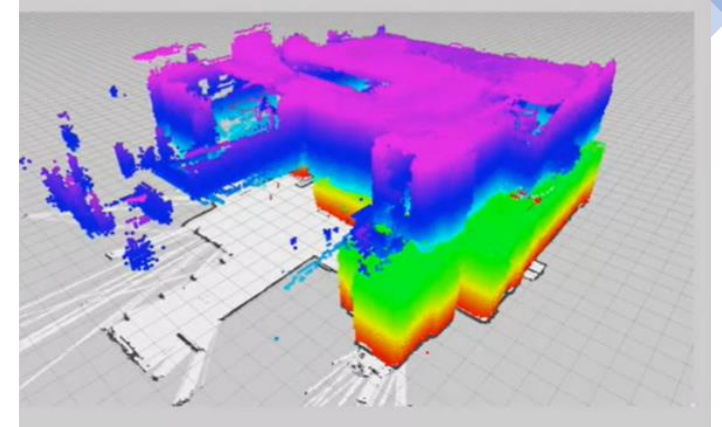
- Wrapping up SLAM maths
- Quadrotor Dynamic and Control

# Autonomous Navigation System

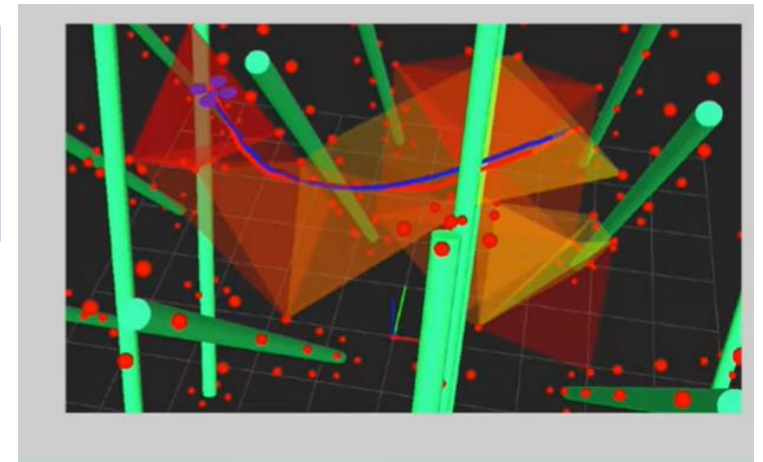
State Estimation

*estimate the position and velocity  
(including rotation and angular  
velocity of the robot)*

Mapping



Planning



Control

*command motors and produce  
desired actions in order to navigate  
to desired state*

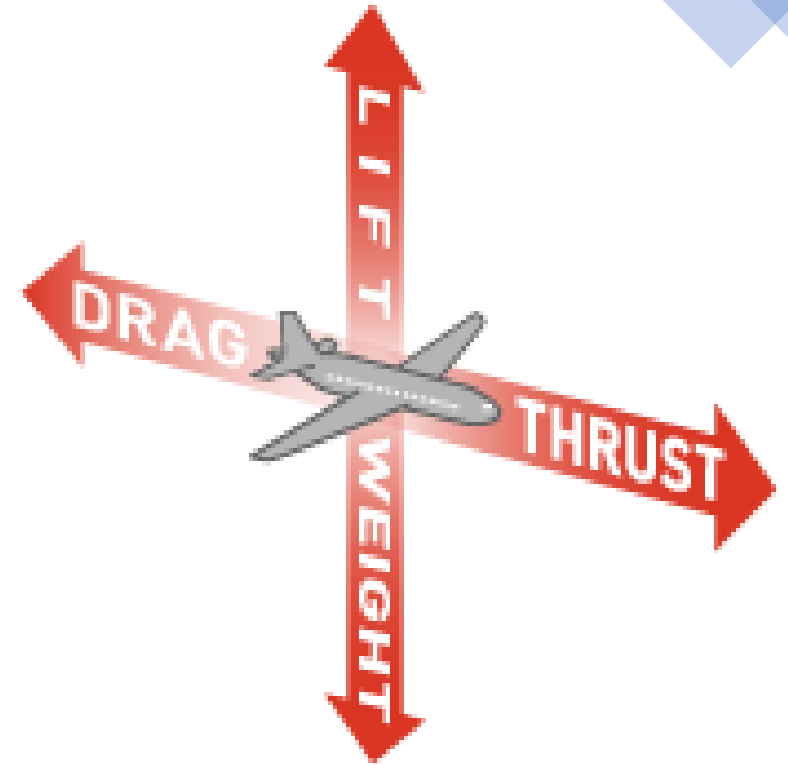
# In this course...

- Quadrotors navigation in simulation
- Race car navigation (mainly final project)



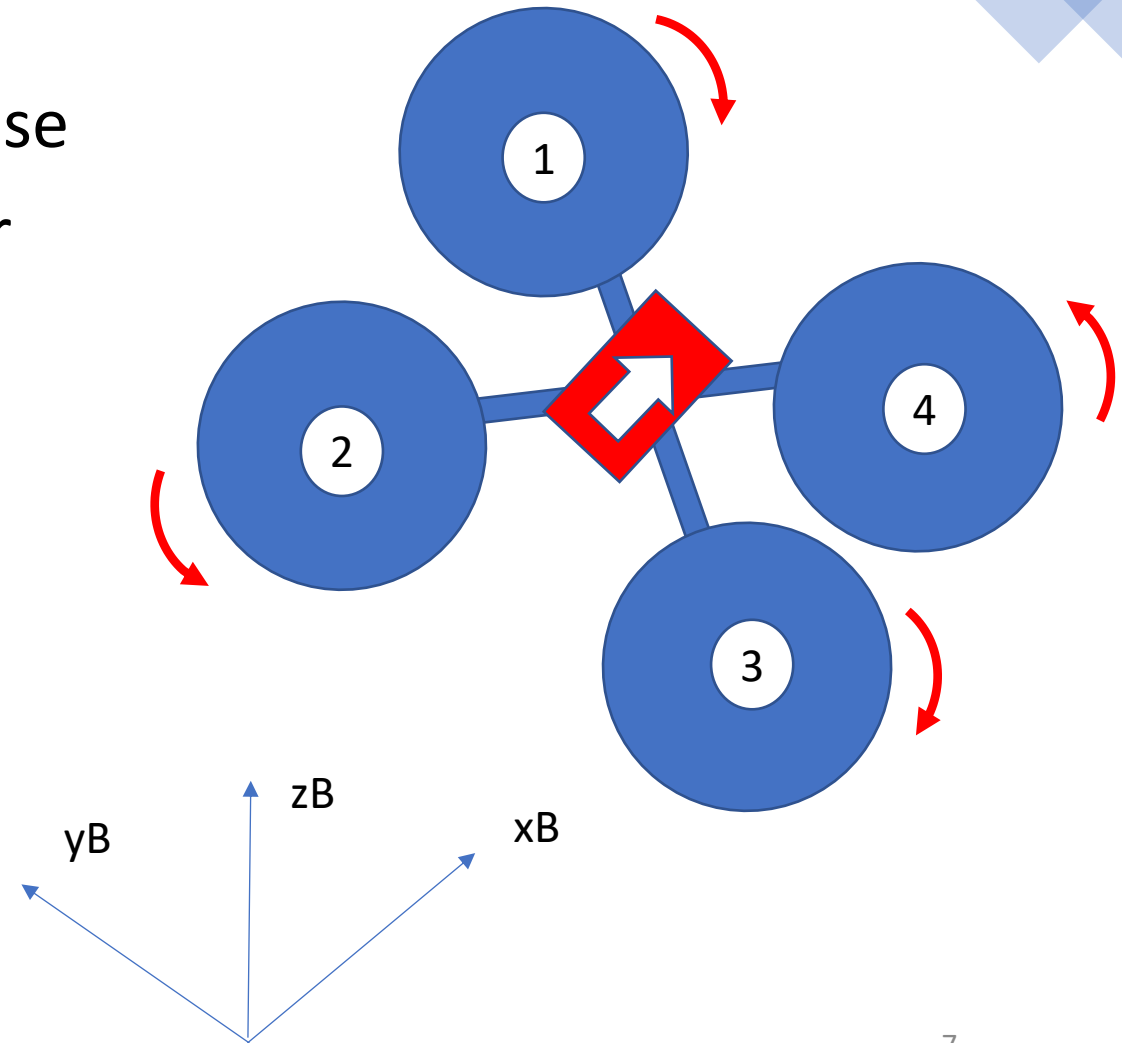
# Aerial Navigation System

1. **Weight** : gravity
2. **Lift** is acted as the opposite of weight
3. **Thrust** forces the quadrotor to move to the desired direction
4. **Drag** is the force that acts opposite to the direction of motion.  
*Drag is caused by friction and differences in air pressure.*



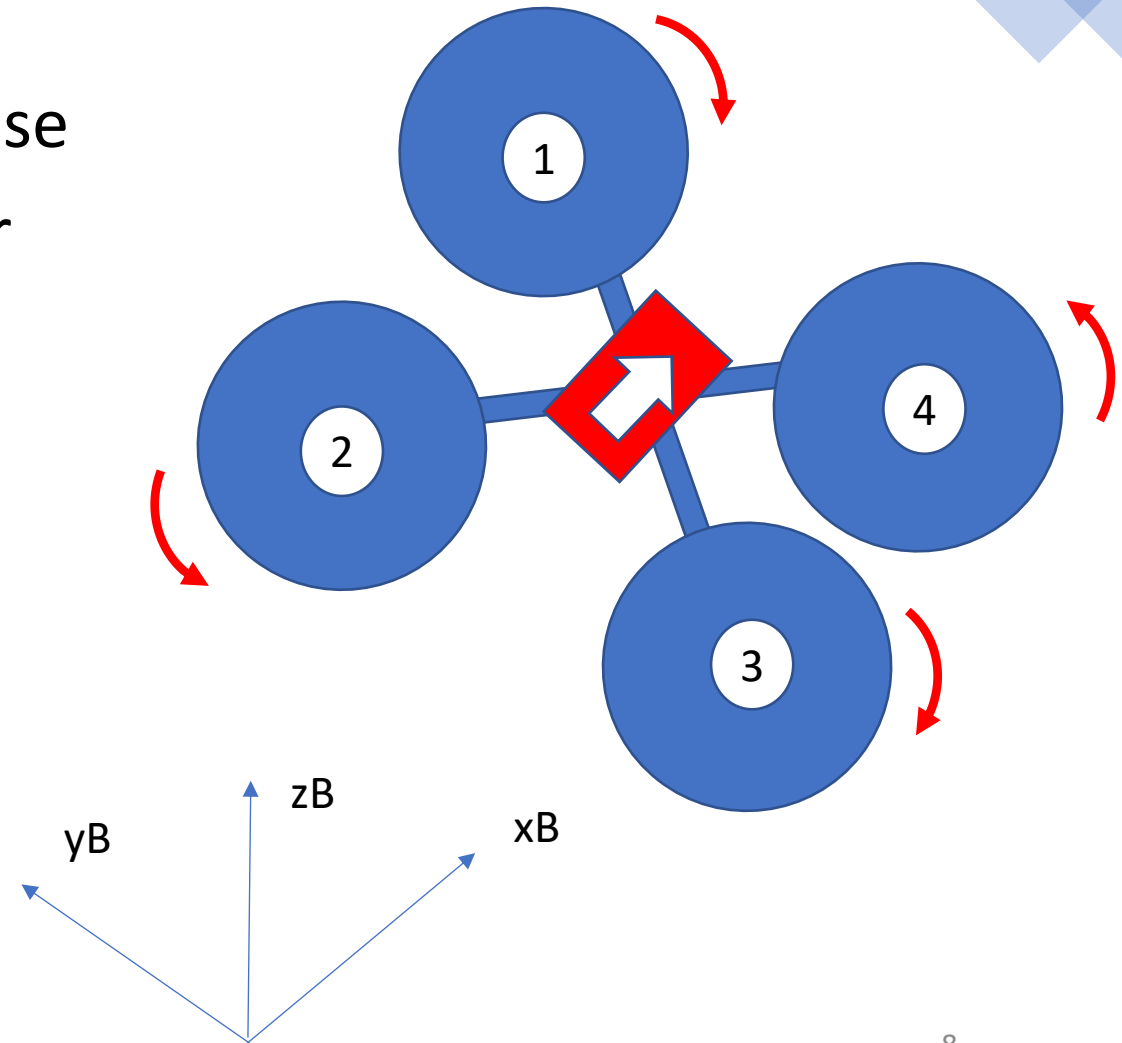
# Quadrotor Dynamics

- Propellers 1 and 3 are rotating clockwise
- Propellers 2 and 4 are rotating counter clockwise
- A quadrotor changes its orientation, depending on the velocity of these propellers



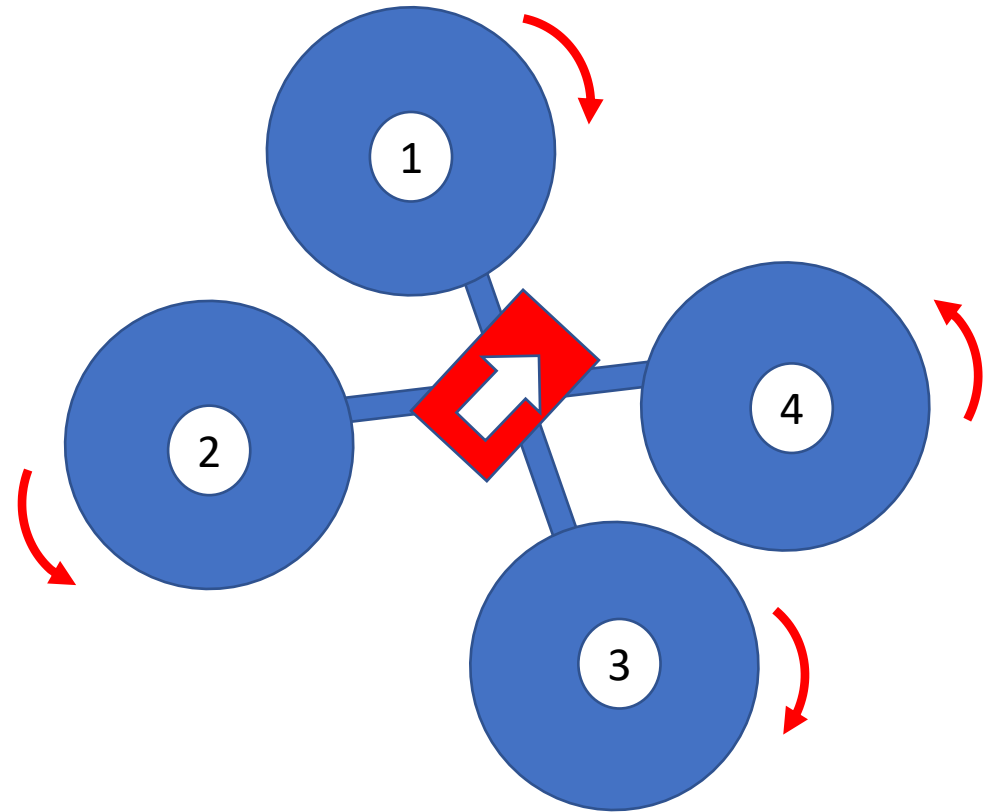
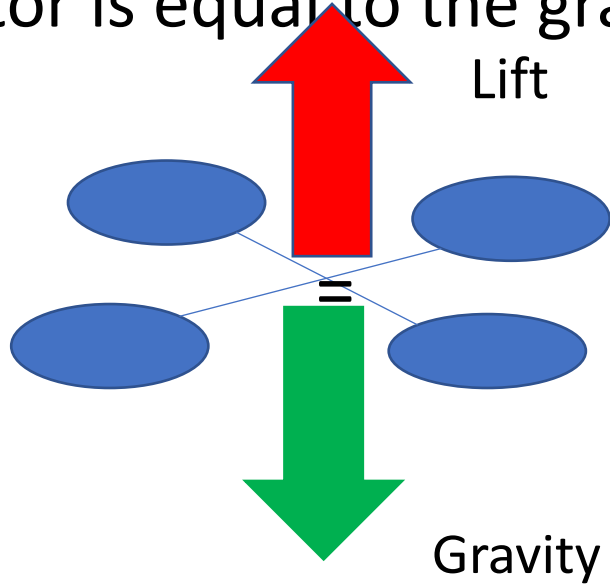
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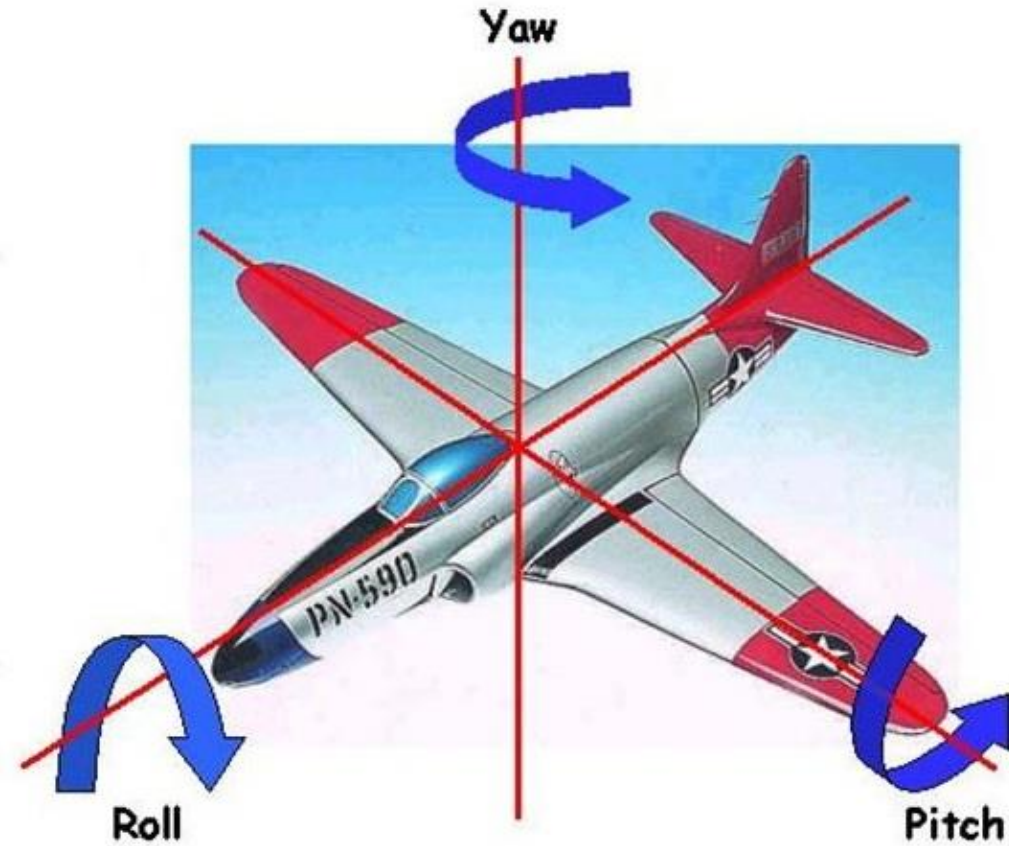
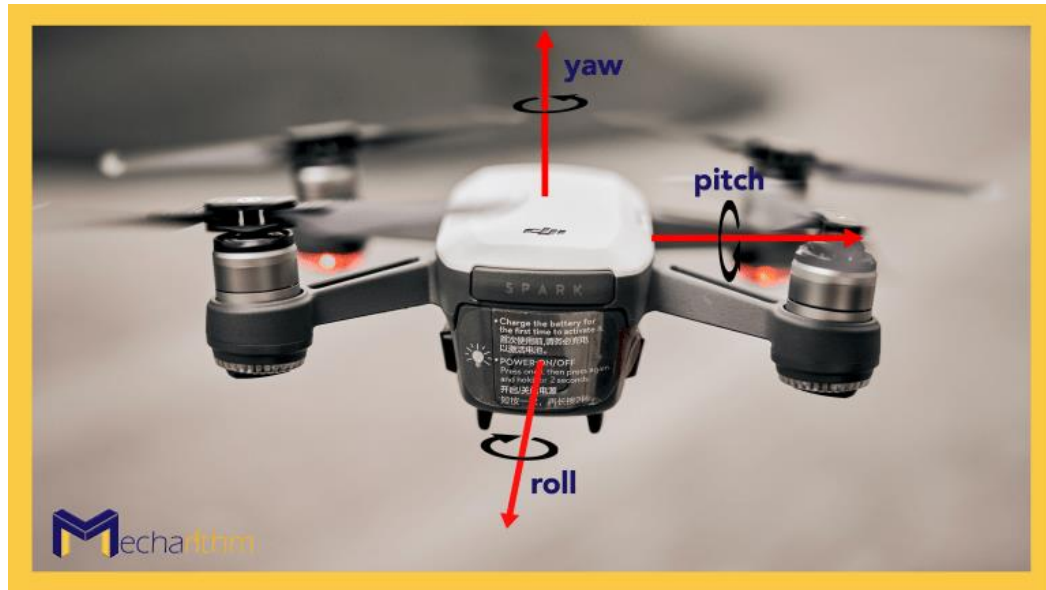


# Hovering

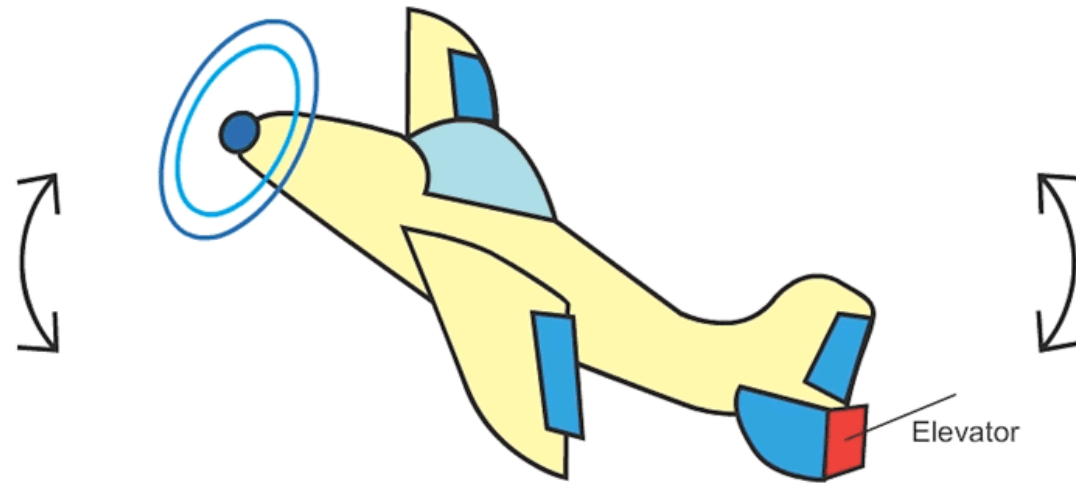
- When the lift force is equal to gravity: Here the total thrust forces in quadrotor is equal to the gravity



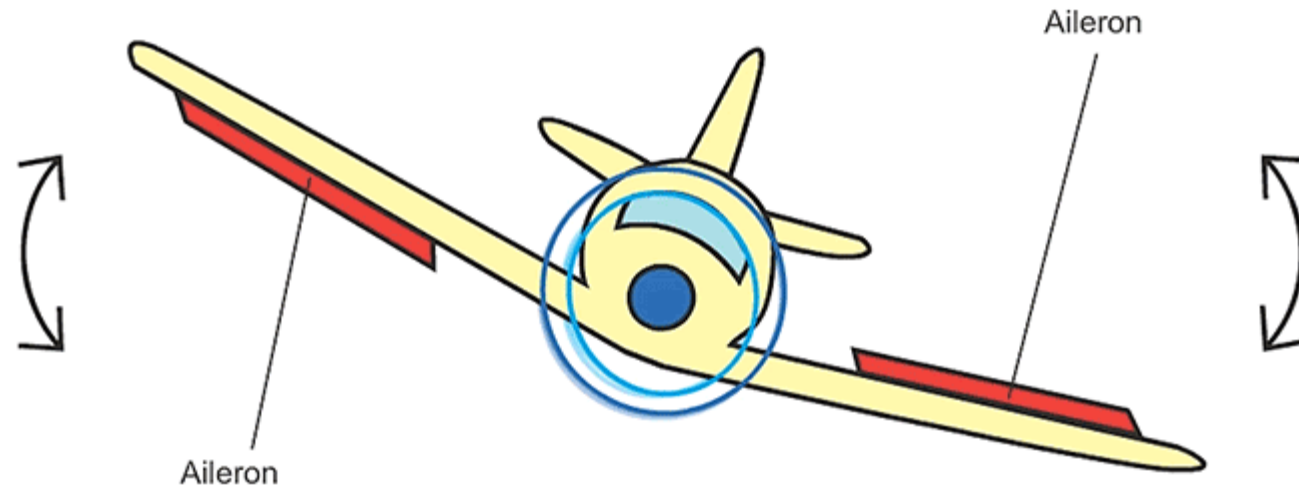
# Pitch/Roll/Yaw



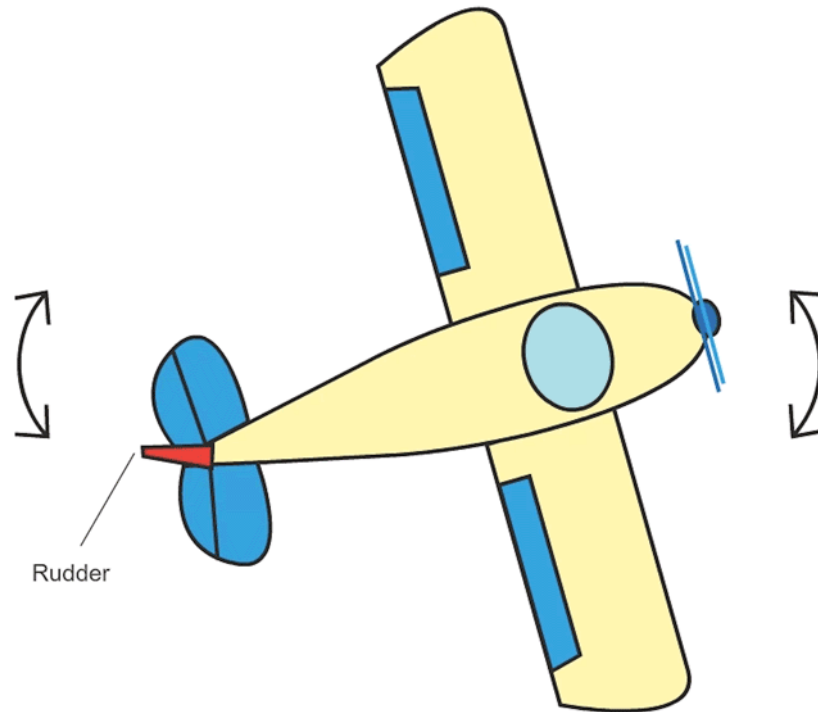
# Pitch!



# Roll!

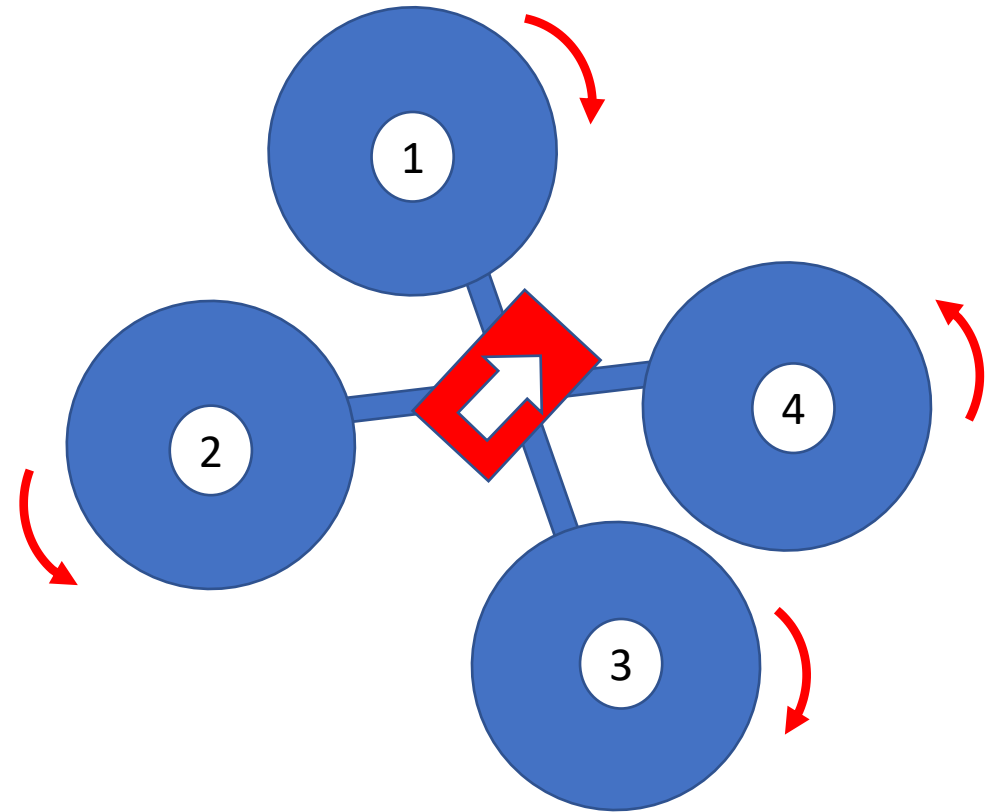
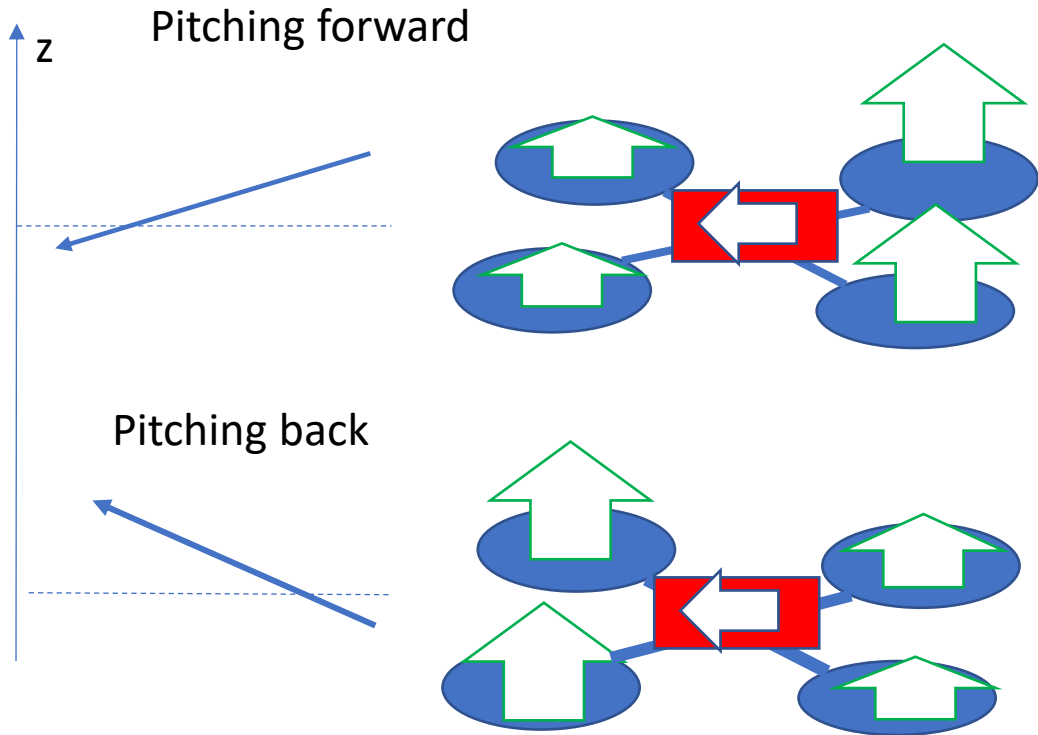


# Yaw!



# Pitching

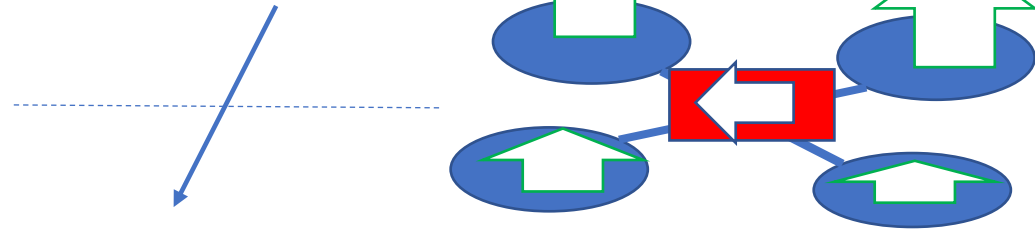
- Motors on the back are rotating faster (top) or slower (bottom) than the motors on the front.



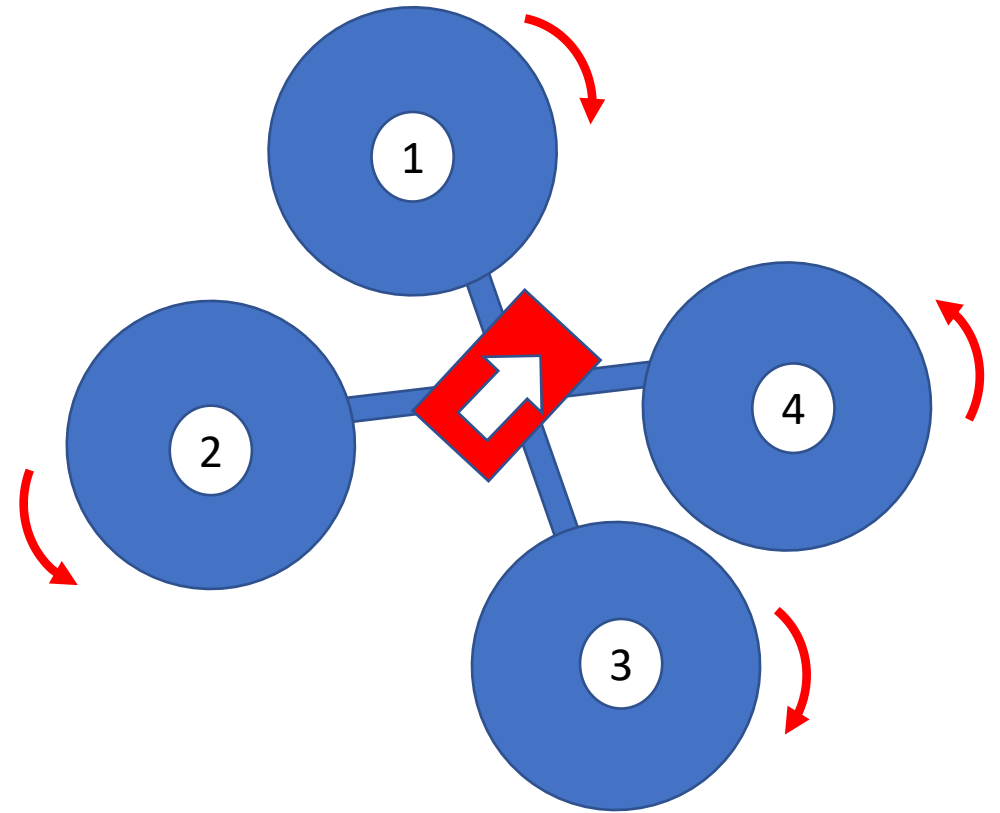
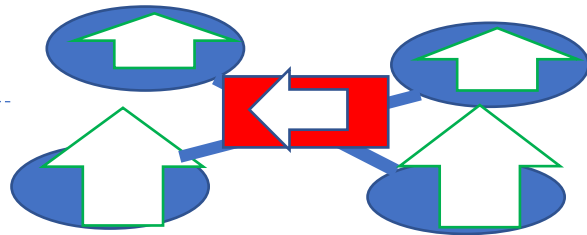
# Rolling

- Motors on the left are rotating faster (top) or slower than motors on the right

Rolling left

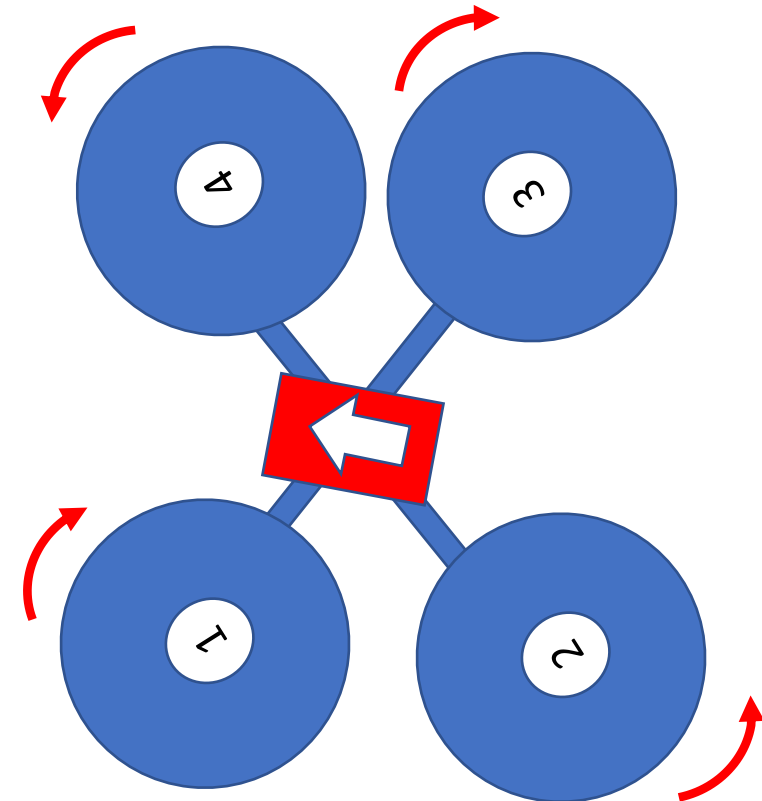
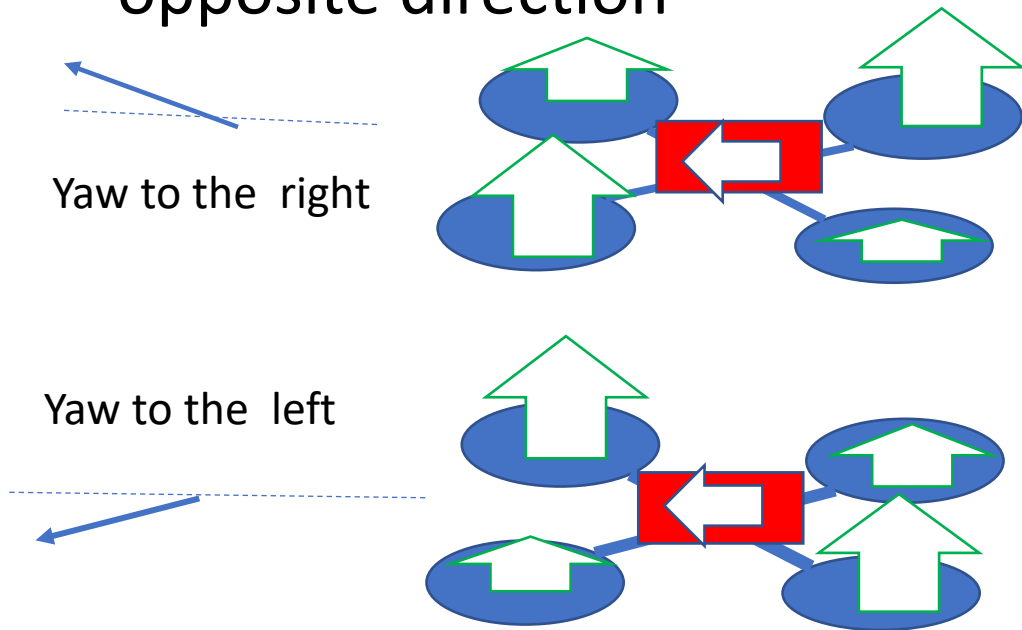


Rolling right

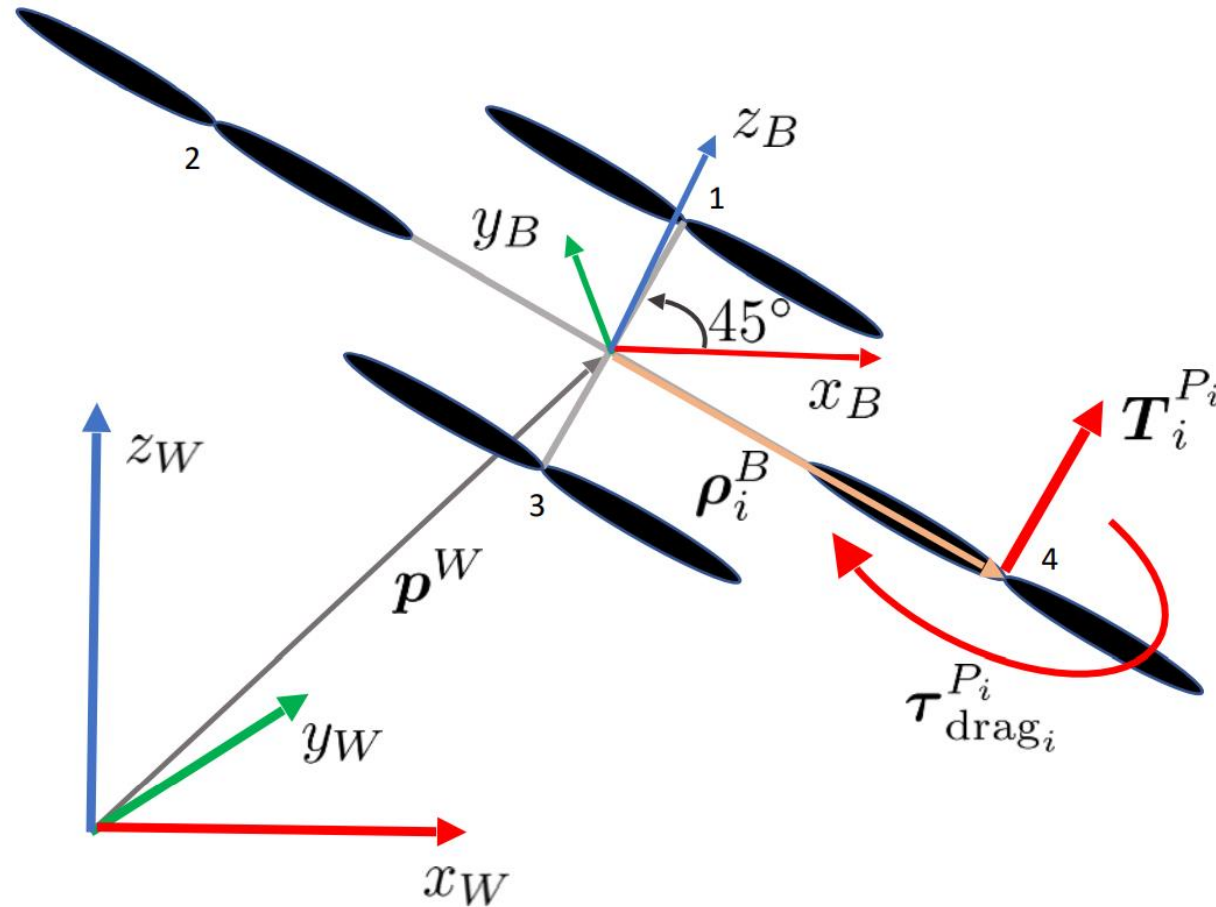


# Yaw

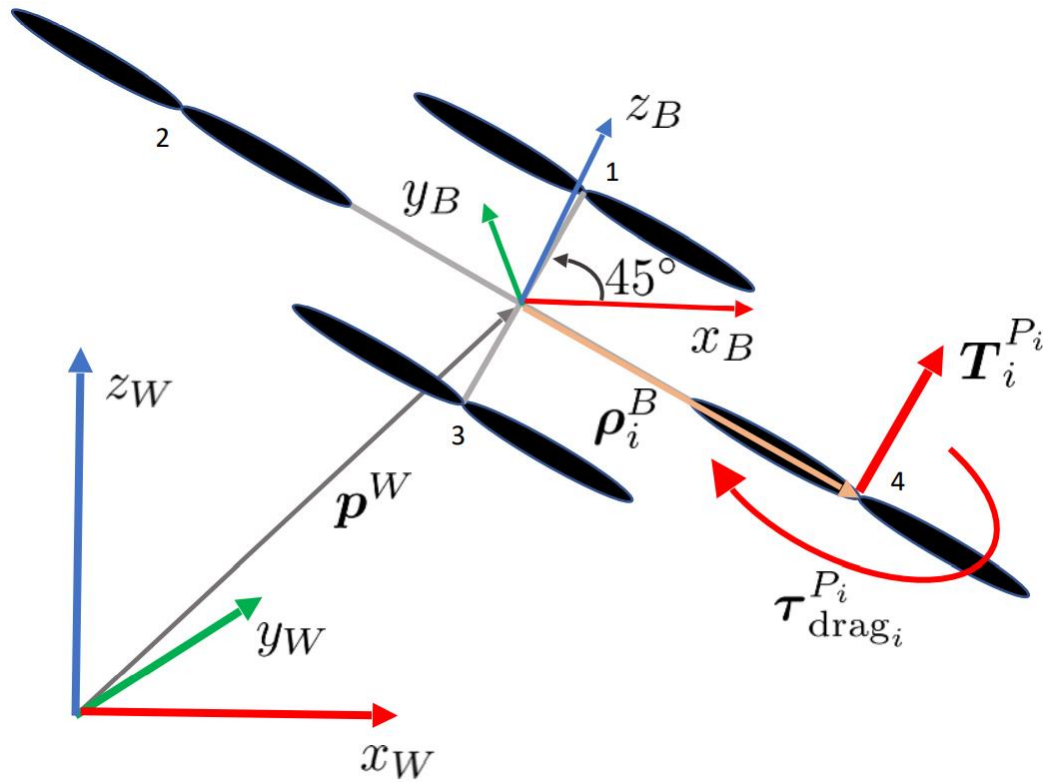
- Motors that rotating counter clockwise are rotating faster (top- yaw to the right) or slower (bottom- yaw to the right) than the motors that rotating clockwise
- Note that based on the 3<sup>rd</sup> Newton's law, the drone is rotating in the opposite direction



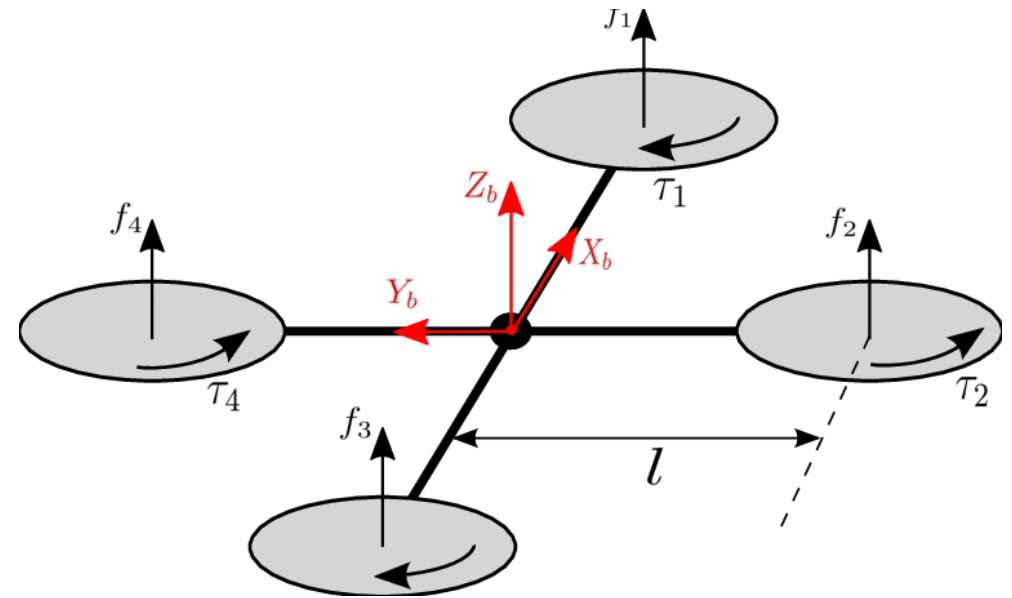
# Quadrotor Dynamics



# Body frames –different settings



This class setting



Another setting

# Quadrotor Model

Let us introduce the Newton-Euler equation, describing the translational and rotational dynamics of a rigid body:

$$\begin{bmatrix} \mathbf{f}^w \\ \boldsymbol{\tau}^B \end{bmatrix} = \begin{bmatrix} m\mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathcal{J} \end{bmatrix} \begin{bmatrix} \mathbf{a}^w \\ \boldsymbol{\alpha}^B \end{bmatrix} + \begin{bmatrix} \mathbf{0} \\ \boldsymbol{\omega}^B \times \mathcal{J} \boldsymbol{\omega}^B \end{bmatrix} \quad (6.1)$$

with symbols as defined in Table 6.1. Note that the translational part is the usual “ $\mathbf{f} = m\mathbf{a}$ ” relation, while the rotational part is slightly more complicated. In particular, the extra cross product term at the end emerges when expressing the torque in the body frame (see [https://en.wikipedia.org/wiki/Euler%27s\\_equations\\_\(rigid\\_body\\_dynamics\)](https://en.wikipedia.org/wiki/Euler%27s_equations_(rigid_body_dynamics)) for a justification).

|                                           |                                                                               |
|-------------------------------------------|-------------------------------------------------------------------------------|
| $\mathbf{f}^w \in \mathbb{R}^3$           | applied total forces expressed in the world frame                             |
| $\boldsymbol{\tau}^B \in \mathbb{R}^3$    | applied total torques expressed in the body frame                             |
| $m \in \mathbb{R}_+$                      | body mass                                                                     |
| $\mathcal{J} \in \mathbb{R}^{3 \times 3}$ | moment of inertia about the center of mass                                    |
| $\mathbf{I}_n$                            | $n \times n$ identity matrix                                                  |
| $\mathbf{a}^w \in \mathbb{R}^3$           | translational acceleration of the center of mass expressed in the world frame |
| $\boldsymbol{\alpha}^B \in \mathbb{R}^3$  | angular acceleration of the body expressed in the body frame                  |
| $\boldsymbol{\omega}^B \in \mathbb{R}^3$  | angular velocity of the body expressed in the body frame                      |

# Considering Gravity...

In the following, we discuss the nature of the forces  $\mathbf{f}^w$  and the torques  $\boldsymbol{\tau}^B$  when the rigid body is a quadrotor. The first external force we consider is *gravity*, which we make explicit in our model and move to the right-hand-side of (6.1) (note:  $\mathbf{g}^w \in \mathbb{R}^3$  is the gravity vector expressed in the world frame):

$$\begin{bmatrix} \mathbf{f}^w \\ \boldsymbol{\tau}^B \end{bmatrix} = \begin{bmatrix} m\mathbf{I}_3 & \mathbf{0} \\ \mathbf{0} & \mathcal{J} \end{bmatrix} \begin{bmatrix} \mathbf{a}^w \\ \boldsymbol{\alpha}^B \end{bmatrix} + \begin{bmatrix} -m\mathbf{g}^w \\ \boldsymbol{\omega}^B \times \mathcal{J}\boldsymbol{\omega}^B \end{bmatrix} \quad (6.2)$$

Other forces and torques are due to aerodynamic effects. The two main aerodynamic effects for a quadrotor are:

- Thrust Force
- Rotor Drag

# Other forces...

- Hub Force
- Hub Moment
- Rolling Moment
- Ground effect

At low speed ( $< 10\text{m/s}$ ), the latter are by more than one order of magnitude smaller than the rotor drag and the trust force, and can therefore be neglected. A more detailed explanation of these effects is given in [\[1\]](#).

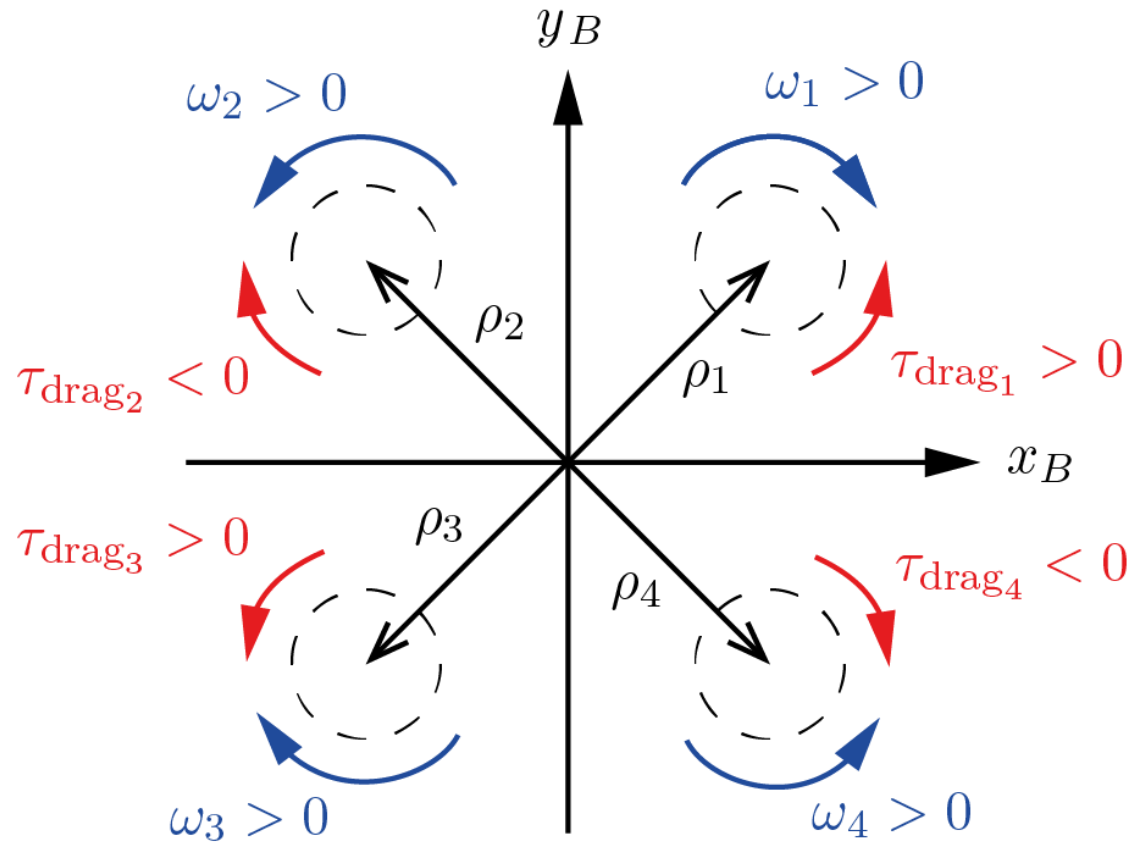
# Forces and Torques..

$$\mathbf{f}_{thrust}^B = \sum_{i=1}^4 \mathbf{R}_{P_i}^B \mathbf{T}_i^{P_i}, \quad (6.3)$$

with  $\mathbf{R}_{P_i}^B$  being the rotation matrix from propeller- to body-frame ( $\mathbf{R}_{P_i}^B = \mathbf{I}_3$  for standard quadrotors).

$$\mathbf{T}_i^{P_i} = c_f w_i |w_i| \mathbf{e}_3,$$

# This course (TESSE simulator) convention



(top view of the quadrotor,  $\|\rho_i\|_2 = d$ )

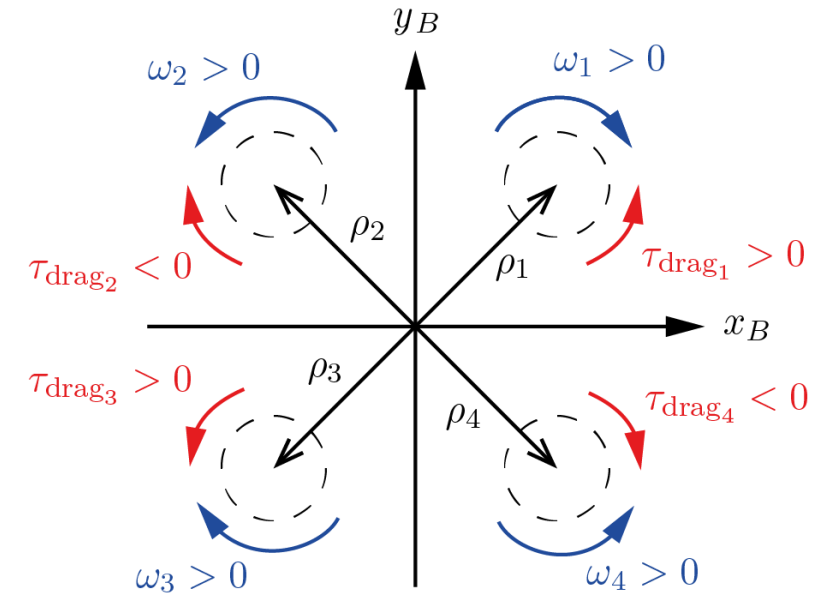
# Forces and Torques..

$$\boldsymbol{\tau}^B = \boldsymbol{\tau}_{drag}^B + \boldsymbol{\tau}_{thrust}^B$$

$$\boldsymbol{\tau}_{thrust}^B = \sum_{i=1}^4 (\boldsymbol{\rho}_i^B \times \mathbf{R}_{P_i}^B \mathbf{T}_i^{P_i}),$$

$$\boldsymbol{\tau}_{drag}^B = \sum_{i=1}^4 \mathbf{R}_{P_i}^B \boldsymbol{\tau}_{drag_i}^{P_i}$$

$$\boldsymbol{\tau}_{drag_i}^{P_i} = (-1)^{(i+1)} c_d w_i |w_i| \mathbf{e}_3$$



(top view of the quadrotor,  $\|\rho_i\|_2 = d$ )

# Finally...

$$\begin{bmatrix} m\mathbf{a}^w \\ \mathcal{J}\boldsymbol{\alpha}^B \end{bmatrix} = \begin{bmatrix} m\mathbf{g}^w \\ -\boldsymbol{\omega}^B \times \mathcal{J}\boldsymbol{\omega}^B \end{bmatrix} + \begin{bmatrix} \mathbf{R}_B^w & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_3 \end{bmatrix} \begin{bmatrix} \mathbf{f}_{thrust}^B \\ \boldsymbol{\tau}_{drag}^B + \boldsymbol{\tau}_{thrust}^B \end{bmatrix} = \begin{bmatrix} -mg\mathbf{e}_3 \\ -\boldsymbol{\omega}^B \times \mathcal{J}\boldsymbol{\omega}^B \end{bmatrix} + \begin{bmatrix} \mathbf{R}_B^w & \mathbf{0} \\ \mathbf{0} & \mathbf{I}_3 \end{bmatrix} \mathbf{F}\mathbf{w} \quad (6.7)$$

where we defined  $\mathbf{w} = [w_1|w_1|, w_2|w_2|, w_3|w_3|, w_4|w_4|]^T$ , we defined a suitable matrix  $\mathbf{F}$ , and we observed that for a typical choice of world frame (with the  $z$  axis pointing upwards) it holds  $\mathbf{g}^w = -g\mathbf{e}_3$ . We remark that the vector  $\mathbf{F}\mathbf{w}$  essentially includes the forces (except gravity) and the torques in the body frame, which using (6.3)-(6.6) and assuming  $\mathbf{R}_{P_i}^B = \mathbf{I}_3$ , can be written as:

$$\begin{bmatrix} f_x^B \\ f_y^B \\ f_z^B \\ \tau_x^B \\ \tau_y^B \\ \tau_z^B \end{bmatrix} = \mathbf{F}\mathbf{w} = \begin{bmatrix} c_f\mathbf{e}_3 & c_f\mathbf{e}_3 & c_f\mathbf{e}_3 & c_f\mathbf{e}_3 \\ c_d\mathbf{e}_3 + c_f\boldsymbol{\rho}_1^B \times \mathbf{e}_3 & -c_d\mathbf{e}_3 + c_f\boldsymbol{\rho}_2^B \times \mathbf{e}_3 & c_d\mathbf{e}_3 + c_f\boldsymbol{\rho}_3^B \times \mathbf{e}_3 & -c_d\mathbf{e}_3 + c_f\boldsymbol{\rho}_4^B \times \mathbf{e}_3 \end{bmatrix} \mathbf{w} \quad (6.8)$$